

Tech & Teach

Digital Chip Design Program

RTL Design

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Sequential Datapath Design Part 2

Version 1.0

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Task #8.1

```

module Counter(
    input logic clk,
    output logic [3:0] Q = 4'b0000
);
always @(posedge clk)
begin
    Q <= Q + 1;
end
endmodule

```

```

`timescale 1ns/1ps
module Counter_tb;
logic clk;
logic [3:0] Q;
Counter dut(
    .clk(clk),
    .Q(Q)
);
always #5 clk = ~clk;

```

```

initial begin

```

```

    clk = 0;

```

```

    #100;

```

```

    $finish;

```

```

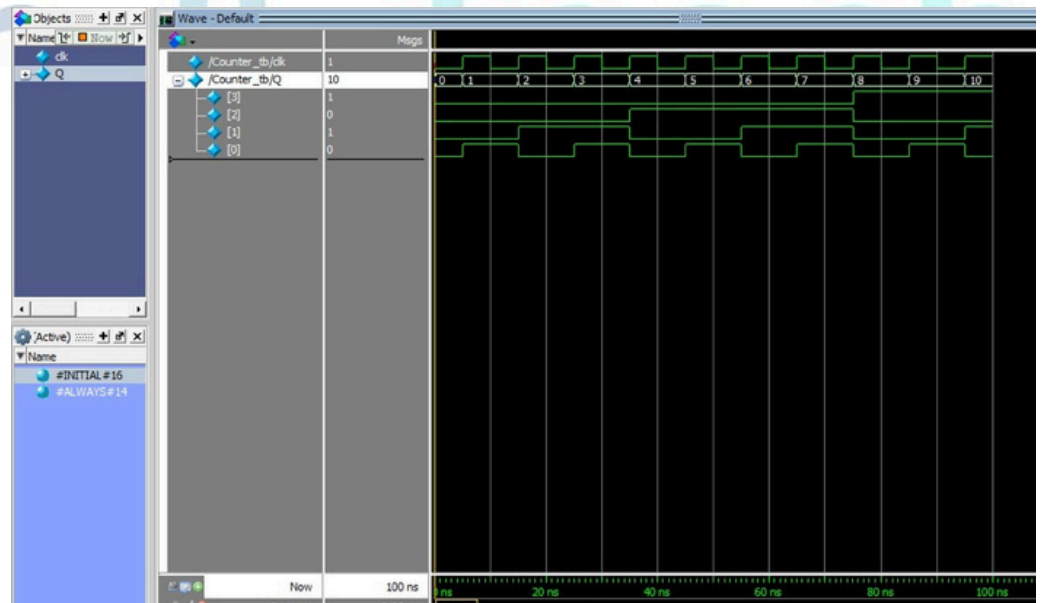
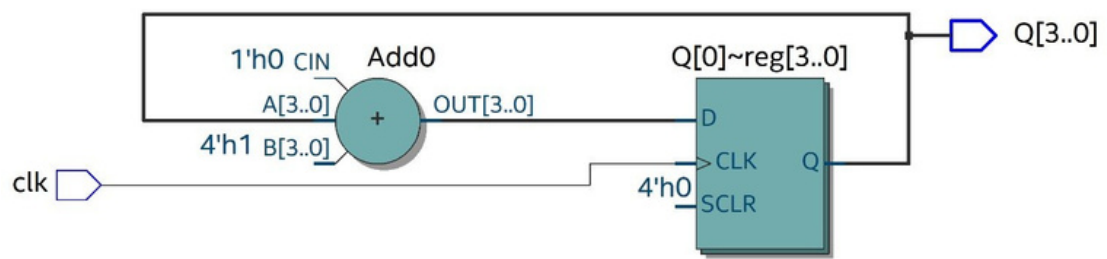
end

```

```

endmodule

```

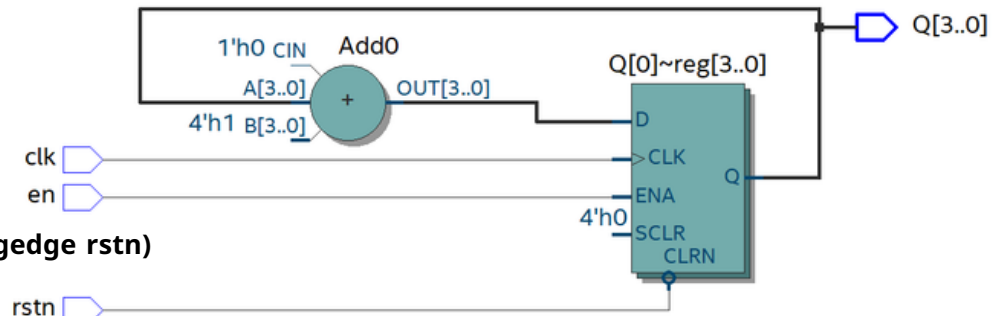


Task #8.2

```

module Counter #(
    parameter N = 4
)(
    input logic clk,
    input logic rstn,
    input logic en,
    output logic [N-1:0] Q
);
always @(posedge clk , negedge rstn)
begin
    if (!rstn)
        Q <= '0;
    else if (en)
        Q <= Q + 1;
    end
endmodule

```



```

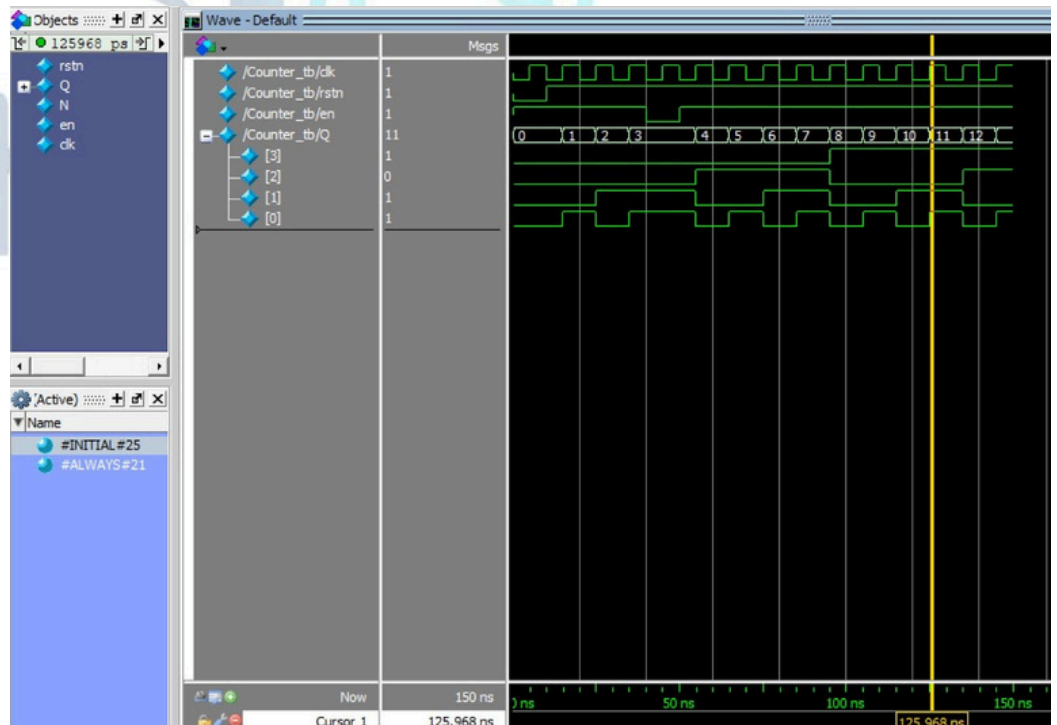
`timescale 1ns / 1ps
module Counter_tb;
parameter N = 4;

```

```

    logic clk;
    logic rstn;
    logic en;
    logic [N-1:0] Q;
    Counter #(
        .N(N)
    ) dut (
        .clk(clk),
        .rstn(rstn),
        .en(en),
        .Q(Q)
    );
    always begin
        #5 clk = ~clk;
    end
    initial begin
        clk = 0;
        rstn = 0;
        en = 1;
        #10 rstn = 1;
        #30 en = 0;
        #10 en = 1;
        #100 $finish;
    end
endmodule

```

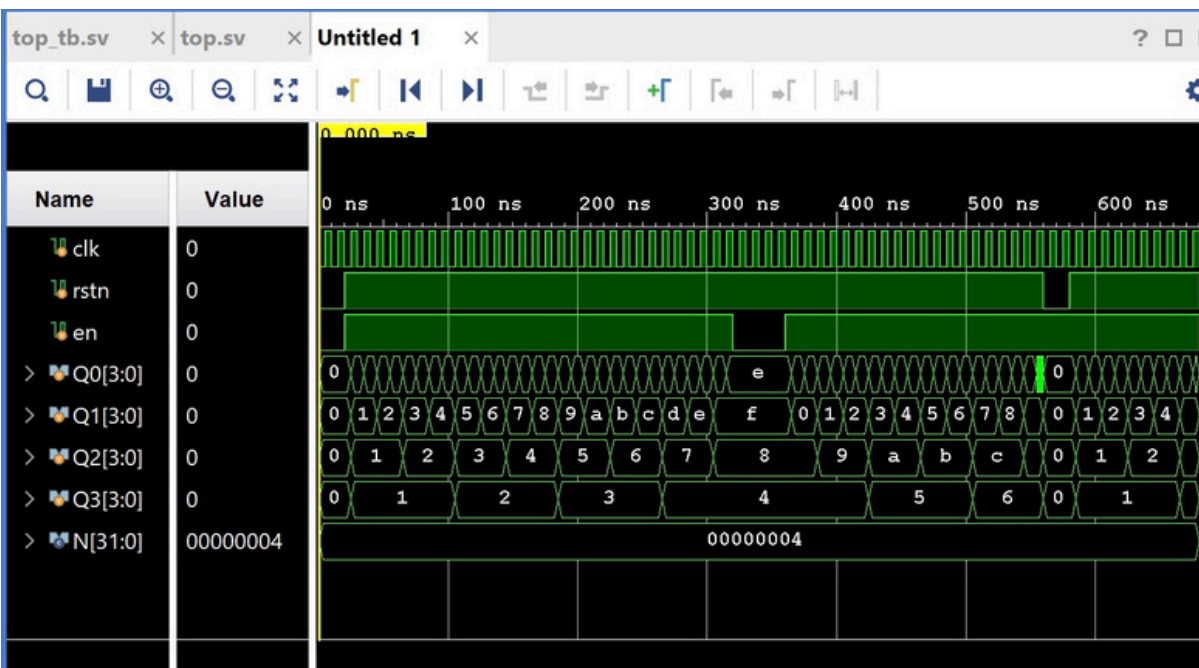


Task #8.3

```
module top #(parameter N = 4)(
  input logic rstn, clk,en,
  output logic [N-1:0] Q);
  logic [N-1:0] Q1,Q2,Q4,Q8;
  logic F2, F4, F8;
  Counter counF(
    .clk(clk),
    .rstn(rstn),
    .en(en),
    .Q(Q1));
  Counter counF2(
    .clk(F2),
    .rstn(rstn),
    .en(en),
    .Q(Q2) );
  Counter counF4(
    .clk(F4),
    .rstn(rstn),
    .en(en),
    .Q(Q4) );
  Counter counF8(
    .clk(F8),
    .rstn(rstn),
    .en(en),
    .Q(Q8)
  );
  clockDivider clkDiv(
    .F(clk),
    .rstn(rstn),
    .F2(F2),
    .F4(F4),
    .F8(F8)
  );
endmodule
```

```
module clockDivider (F, rstn, F2, F4, F8);
  input logic F, rstn;
  output logic F2, F4, F8;
  always @(posedge F, negedge rstn)
    if(rstn) begin
      F2 <= ~F2;
    end
    else F2 = 0;
  always @(posedge F2, negedge rstn)
    if(rstn) begin
      F4 <= ~F4;
    end
    else F4 = 0;
  always @(posedge F4, negedge rstn)
    if(rstn) begin
      F8 <= ~F8;
    end
    else F8 = 0;
endmodule
```

```
`timescale 1ns/1ps
module top_tb;
  parameter N = 4;
  logic clk;
  logic rstn;
  logic en;
  logic [N-1:0] Q;
  top #(N) DUT (
    .clk(clk),
    .rstn(rstn),
    .en(en),
    .Q(Q));
  initial begin
    clk = 0;
    #5 clk = ~clk;
  end;
  initial begin
    rstn = 0;
    en = 0;
    rstn = 1;
    en = 1;
    en = 0;
    en = 1;
    rstn = 0;
    rstn = 1;
    $finish;
  end;
endmodule
```



Work distribution

Section	Almas	Atheer	Mjd	Maha	Jood
Report	✓	✓	✓	✓	✓
Task8.1	✓	✓	✓	✓	✓
Task8.2	✓	✓	✓	✓	✓
Task8.3	✓	✓	✓	✓	✓

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